

Elen Yeghiazaryan

Location: Armenia

Email: elenyeg005@gmail.com

LinkedIn: [linkedin.com/in/elen-yeghiazaryan](https://www.linkedin.com/in/elen-yeghiazaryan)

GitHub: github.com/elenyeghiazaryan

About Elen

Purposeful and inquisitive student with a strong passion for engineering and technology. Eager to solve real-world problems at the intersection of engineering, computation, mathematics, and programming. Committed to continuous learning, embracing challenges, and driving personal and professional growth.

Education

- UFAR (French University in Armenia) – Informatics and Applied Mathematics – Sep 2023 – May 2027
- Université Paul Sabatier Toulouse III – Informatics and Applied Mathematics – Sep 2024 – May 2027
- Synopsys Armenia Educational Department – Semiconductor Engineering – 2024 – 2026
- Heratsi High School – Economics Stream – Sep 2020 – May 2023
- TUMO Labs – ClimateNet – Mar 2026 – Present

Work Experience

ESL Teacher – Academy Polyglot (2024 – 2025)

- Designed lessons for A1–C1 learners
- Created grammar, vocabulary, and speaking activities
- Developed interactive learning games

- Delivered engaging online and in-person lessons
- Supported students individually to improve fluency and accuracy
- Improved students' speaking confidence and pronunciation
- Prepared students for exams and presentations

Volunteer Experience

- **Armenian Red Cross Society** – Community Volunteer – (Jul 2021 – Nov 2021)

Helped children with homework and led extracurricular activities. Involved them in clubs (math, reading, English, sports, etc.) to build communication skills and broaden their knowledge.

- **Ministry of High-Tech Industry of the Republic of Armenia** – Call Center Operator – (Mar 2022 · 1 month)

Conducted calls and surveys with participants of ministry-organized courses to assess satisfaction, productivity, and the extent to which the courses helped them find employment.

- **Career Center** – Community Development Volunteer – (Jun 2022)

Completed a first aid course and then educated other students on first aid techniques. Focused on raising awareness and encouraging knowledge sharing among peers.

- **PAVU** – Volunteer – (Oct 2025)

Attended the Digitec technology event, engaged with industry professionals, and explored emerging tech trends.

Certificates

- TOEFL
- DELF B2
- Technovation Girls
- Tumo

Technical Skills

Programming Languages - Python, C, Java

Core CS & Engineering - Networking, Systems, Statistics, Numerical Analysis, Machine Learning, Computer Architecture, Microelectronics / Semiconductor Engineering, Object-Oriented Programming, IoT, Data Structures & Algorithms, Databases (MySQL)

Tools & Software

- Systems & IoT: Raspberry Pi, ESP32, Arduino
- Version Control & Editors: Git (GitHub), Vim, Jupyter Notebook, Google Colab, Visual Studio, VS Code, IntelliJ IDEA
- Databases: MySQL (querying, administration)
- Networking: TCP/IP, DNS, DHCP, VLANs; ping, traceroute, netstat
- Office & Productivity: Microsoft Office (Word, Excel, PowerPoint, Outlook), Microsoft Teams, OneDrive
- Operating Systems & Shells: Linux (Ubuntu), macOS (terminal), Windows, Bash / Zsh

Languages

- Armenian – Native
- English – Native
- French – Advanced
- Russian – Conversational
- German – Elementary

Projects

1. Raspberry Pi Mini Network & Gateway

Turned a Raspberry Pi into a network gateway and virtual access point despite a single-chip interface constraint. Handled static routing, DHCP (pool

192.168.50.10–100), local DNS, packet forwarding, and traffic bridging for client devices – all routed through dnsmasq/iptables and accessible via `iot.local`.

GitHub: [TUMO-Labs/aem_wifi](https://github.com/TUMO-Labs/aem_wifi)

2. Smart Evacuation System

A B2B safety platform that computes real-time personalised evacuation routes for large multi-storey buildings using BLE beacon positioning and graph-based BFS pathfinding – backed by a fully normalised SQL Server database.

GitHub: [elenyeghiazaryan/Smart-Evacuation-System](https://github.com/elenyeghiazaryan/Smart-Evacuation-System)

3. Global Demographic Structure Analysis

Comprehensive statistical analysis of demographic factors to uncover patterns essential for infrastructure, healthcare, and economic planning. Uses a multi-stage data pipeline (web scraping → descriptive statistics → hypothesis testing → advanced visualizations).

GitHub: [elenyeghiazaryan/Global-Demographic-Structure-Analysis](https://github.com/elenyeghiazaryan/Global-Demographic-Structure-Analysis)

4. Voting App

Console-based Python voting system where users can vote and be voted for. Includes an admin panel to track and manage votes.

GitHub: [elenyeghiazaryan/voting_app](https://github.com/elenyeghiazaryan/voting_app)

5. Serverless Cloud APIs

Built serverless backend APIs with AWS Lambda, S3, and API Gateway. Configured custom routing, method parsing, and structured JSON logs.

GitHub: [TUMO-Labs/aws-tasks](https://github.com/TUMO-Labs/aws-tasks)

6. RaspberryChat

Simple real-time chat app built with Flask and Flask-SocketIO. Deployed on a Raspberry Pi using Nginx and Gunicorn.

GitHub: [TUMO-Labs/raspberryChat](https://github.com/TUMO-Labs/raspberryChat)

7. Design Patterns

Implemented classic object-oriented design patterns in Python, including Abstract Factory, Adapter, Builder, Chain of Responsibility, Decorator, Facade, Factory, and

Singleton. Each pattern is a standalone, runnable module with real-world inspired examples.

GitHub: [elenyeghiazaryan/design-patterns-python](https://github.com/elenyeghiazaryan/design-patterns-python)